

Make It in China — HSK Conversation Game: Design Doc

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Summary

A browser game in three.js where the player arrives in China knowing a few words and has to earn a living. Every job, purchase and conversation runs in Mandarin. Understanding more words unlocks better jobs, so language is the player's earning power.

- **Teaches:** HSK 2.0 vocabulary and sentence patterns, one HSK level per phase (Phase 1 = HSK 1, up to Phase 4 = HSK 4).
- **Feels like:** Abeto's *Messenger* — calm, small world, talk-and-fetch quests, toon shading with outlines.
- **Looks like:** a Game Boy Advance-era fixed-camera game — small characters seen from a high angle, simple levels.
- **Build now:** Phase 1 only. One street, three jobs, 150 words.
- **Decide after:** whether testers finish it, remember the words, and ask for more. Phase 2 is built only if they do.

"Make It in China" is a working title.

Concept and core loop

The player is taught a handful of words, then left in a Chinese city to live and earn. Each new word arrives inside a new situation, and the same word comes back later with a different nuance. Money gives the player a reason to care whether they understood.

flowchart LR

```

A[Wake up<br/>new day] --> B[Pick a job<br/>or errand]
B --> C[NPC speaks<br/>audio + hanzi]
C --> D[Player replies]
D -->|understood| E[Earn yuan<br/>word logged]
D -->|misunderstood| F[Funny mix-up<br/>small cost]
F --> C
E --> G[Spend: rent,<br/>food, tools]
G --> H[New job or<br/>area unlocks]
H --> A

```

Comprehension gates progress. A wrong answer never shows a red cross; the NPC acts on what the player said, and the player sees the result.

Examples of the word-then-nuance pattern:

- 要 first means "want" at a food stall, later "need" from a boss, later "going to" in a schedule.
- 几 vs 多少 — asking "how many" for small and large numbers, learned by getting a delivery count wrong.
- 你 vs 您 — the same sentence lands differently with a shopkeeper and with an employer.
- Measure words at the market: 一个, 一杯, 一本.

Design principles

These are agreed and apply to every scene and system.

1. **Language is the mechanic.** Progress depends on understanding. No quiz screens bolted onto a walking game.
2. **Failure is cheap and funny.** Wrong order delivered, overpaid for cabbage, got off at the wrong stop. Small yuan loss, a laugh, try again. Never game over.
3. **90% known, 1-2 new.** Each exchange uses about 90% words the player has already met, plus one or two new ones. A scene of 4-6 exchanges introduces about 6 words in total. More than that becomes guessing.
4. **Audio first.** The NPC speaks; hanzi are shown; pinyin stays hidden until tapped.
5. **Characters live in the world.** Signs, menus, receipts, price tags and phone messages are all in hanzi. This is the reading practice.

6. **Nuance after experience.** A mentor NPC explains a usage point after the player has run into it, not before.
7. **Hidden spaced repetition.** Words the player missed come back in later scenes. The player never sees a review queue.
8. **Scripted dialogue, not a live LLM.** Keeps vocabulary strictly in-level, runs offline, costs nothing per player. LLM free-chat is a possible later add-on, constrained to the word list.
9. **Time moves in days, not a real-time clock.** Nobody is rushed mid-sentence.
10. **Grey boxes before polish.** Fun and learning get proven with placeholder art. Art comes last.

Content constraints

These are hard requirements from the owner, not style preferences.

No depicted faces

- Every human and animal character has no facial features: no eyes, nose or mouth, on the mesh or in the texture.
- No character portraits in dialogue. Use a name label and a speech bubble.
- Characters are told apart by outfit, colour, headwear, build and what they carry.
- The fixed high camera helps: characters are small and read by silhouette.
- If a downloaded model has a painted face, erase it in the texture atlas before use. If the face is modelled geometry, flatten it in Blender.
- The exact standard (blank head acceptable, or something stricter) is an open question below. Build character loading so the head mesh can be swapped in one place.

No real people or licensed IP

- Jackie Chan Adventures and Messenger are references for look and feel only. No names, likenesses, characters, assets or music from either.
- Do not copy assets out of the Messenger site or its clones.

Economy and theme limits

- Money comes from wages and trade only. No loans, interest, credit or investment returns anywhere in the economy.
- No gambling, lottery or chance-based money mechanics.
- No alcohol. Business meals happen over tea and food.

- No romance plots or suggestive content.
- No magic or supernatural story elements.
- HSK 4 includes 信用卡 (credit card). Teach it as a word seen at a shop counter; keep it out of the player's economy.

Phase plan

Phase N teaches HSK N (HSK 2.0 word lists). Finishing a phase means the player has met every word on that level's list. Each phase is a new district of the same city and a step up the career ladder.

Phase	HSK level	New words (total)	Career stage	What the language test becomes	Reply mechanic
1	HSK 1	150 (150)	Porter, dishwasher, delivery	Did you catch the number, item, place?	Pick a response
2	HSK 2	150 (300)	Waiter, shop assistant	Orders, prices, time, directions, bargaining	Assemble word tiles
3	HSK 3	300 (600)	Own stall or small shop	Suppliers, hiring, complaints	Type pinyin
4	HSK 4	600 (1,200)	Real company: staff, ads, competitors, travel	Did you say it the right way? Tone, connectives, written messages	Type pinyin; pinyin hints off by default

Notes for later phases:

- Phase 4 shifts from comprehension to register. Several replies are grammatical but differ in tone: how to refuse, apologise to a customer, or criticise an employee.
- Connectives carry the meaning in Phase 4 (即使...也, 不管...都, 既然, 否则). Misreading one flips the terms of a deal, which makes a good cheap failure.
- HSK lists are thin on trade vocabulary in Phases 1–3. A small set of off-list words is allowed (for example 老板, 进货). Mark them as bonus words in the notebook so the HSK mapping stays honest.
- Content volume doubles at Phase 4. Writing, not code, becomes the bottleneck. This is why the dialogue pipeline in the architecture section matters from day one.

Only Phase 1 is in scope now. The engine should assume four districts exist.

Phase 1 scope

Phase 1 covers arrival day through the first week: one street, 150 HSK 1 words, about 25 scenes of 4–6 exchanges each (roughly 6 new words per scene). A tester should finish it in two to three hours across a few sittings.

In scope

- One street with 6–8 locations: rented room, noodle shop, fruit stall, small supermarket, bus stop, warehouse, tea house, a locked gate to the Phase 2 district.
- About 8 NPCs: landlord, mentor (a friendly neighbour), three employers, two shopkeepers, one customer type that repeats.
- Three jobs:
 - **Porter** at the warehouse — numbers, 这/那, 个, 大/小, 多少.
 - **Dishwasher** at the noodle shop — food and drink words, 杯, 碗 (bonus), 不, 有/没有.
 - **Delivery** on foot — places, 在, 哪儿, 前面/后面, names and times.
- Daily-life scenes: paying rent, buying food, asking the time, a phone message from the landlord, a wrong-bus mix-up.
- Wallet, day cycle, notebook, mentor explanations, hidden repetition of missed words.
- Reply mechanic: pick one of 2–4 responses. Pinyin on tap.
- Placeholder art first; final art pass at the end (see Build order).
- Desktop and mobile browser, touch and keyboard.

Out of scope for Phase 1

- Phases 2–4 content, word tiles, typed pinyin input.
- Multiplayer, accounts, leaderboards.
- LLM-driven free conversation.
- Speech recognition.
- A second language of instruction (English glosses only for now).

Deliverable before any code: a scene list — every scene with its location, NPC, job or errand, known words used, and the new words each exchange introduces — so that all 150 words are covered and each appears at least three times.

Learning system

The game tracks every word the player has met and how well they know it, and uses that to choose and adapt scenes. None of this is shown as a study interface.

How a line is presented

- Audio plays first. Hanzi appear in the speech bubble.
- Tap any word: pinyin and English gloss pop up. Tapping is logged as "needed help" for that word.
- A replay button repeats the audio. A slow-speed replay is available.
- New words in a line are subtly underlined the first time they appear.

How the player replies (Phase 1)

- 2-4 response options in hanzi, each with audio on tap.
- Wrong options are plausible, not silly: same word class or sub-theme (三 vs 四 vs 十, not 三 vs 猫).
- Every option leads somewhere. A wrong choice triggers a consequence scene, then a natural retry.

Word state

Each word has a state per player: `unseen → met → shaky → known`. Evidence that moves it: correct reply depending on that word (up), wrong reply or pinyin tap (down), time since last seen (decays toward shaky).

Hidden repetition

- Scenes have slots where a word can be swapped: the customer's order, the number of boxes, the delivery address. The engine fills slots with the player's shaky words.
- Repeatable jobs (one more shift at the warehouse) are the review sessions. They pay a little and draw mostly from shaky words.
- Story scenes unlock only when the words they depend on are at least `met`.

Notebook

- Fills automatically: word, pinyin, gloss, the sentence it was first heard in, audio.
- Grouped by where the player learned it (noodle shop, warehouse), not alphabetically.
- Bonus (off-list) words are marked as such.
- Shows progress toward the HSK list for the phase: for example 87 of 150.

Mentor

A neighbour NPC the player can visit in the evening. After the player has hit a usage point in the day (几 vs 多少, 不 vs 没), the mentor explains it in two or three lines with examples. Explanations are in English in Phase 1.

Failure handling

- A misunderstanding costs a few yuan or a little time, never progress.
- The consequence is shown, not told: the wrong dish arrives, the boxes go to the wrong door.
- After two misses on the same exchange, the NPC rephrases more simply or gestures, so nobody gets stuck.

Economy

The economy exists to make understanding matter, not to be a management sim. Keep it small enough to balance in a spreadsheet.

- **Currency:** yuan (元, shown as 块 in speech). Prices use realistic small numbers so number listening is constant practice.
- **Income:** wages per completed task. A shift is 3–5 exchanges. Better understanding means fewer mix-ups, so more pay per shift.
- **Costs:** weekly rent, daily food, occasional items (bus fare, a phone top-up, a work tool that unlocks a job).
- **Day cycle:** each day has a fixed number of action slots (for example 4). A job shift, an errand or a mentor visit each use one. Sleeping ends the day. No real-time clock.
- **Rent pressure is soft.** Short on rent: the landlord gives a few more days and the player takes extra shifts. No eviction, no debt, no interest.
- **Progression gate:** the Phase 2 district opens when the player has saved a set amount and met all 150 words. Both conditions, so nobody grinds past the language.
- **Failure costs:** a mix-up loses 1–5 yuan or one action slot. Never more.
- **Excluded by rule:** loans, interest, credit, investment returns, gambling. See Content constraints.

Tuning target for Phase 1: a player who understands most things reaches the gate in about 7 in-game days; a struggling player in about 10. Nobody should fall behind permanently.

Visual direction and assets

The look is low-poly models with flat colours, toon shading and dark outlines, seen from a fixed high camera. Most of the Messenger feel comes from shader and colour choices, not from the meshes, so free CC0 models are enough.

Look

- Flat colour materials, no detailed textures. Limited palette per district; warm and calm.
- Toon shading (`MeshToonMaterial` with a 2-3 step gradient map) plus outlines (`OutlineEffect` or an inverted-hull pass). Test both on mobile for cost.
- Fixed camera, high three-quarter angle, slight follow. No player camera control in Phase 1.
- Characters about 1/8 of screen height. Readable by silhouette and colour.
- Hanzi signage everywhere: shop names, menus, price tags, bus stop boards. Render as textures or troika-three-text with a bundled CJK font subset.
- UI: speech bubbles anchored to characters, name labels, wallet and day counter, notebook button. No portraits.

Characters (faceless)

Source	What it gives	Licence	Work needed
Quaternius Universal Animation Library	Blank-headed mannequin + locomotion, idle, carry, emote animations; free Standard tier has 45 clips	CC0	None for face; recolour per character
Universal Animation Library 2	130+ more clips incl. civilian activities (farming, drinking, fishing)	CC0	Pick the civilian clips
Quaternius Universal Base Characters	Base bodies compatible with the library, FBX and glTF	CC0	Check for face geometry; flatten if present

Source	What it gives	Licence	Work needed
Mixamo X Bot / Y Bot	Faceless mannequins + large animation catalogue	Free with Adobe account; check terms before shipping	Convert FBX to GLB
KayKit and Kenney characters	More personality, varied outfits	CC0	Erase painted eyes in the texture atlas

Recommendation: one shared skeleton (the Quaternius rig) for every human character, so all animations work on everyone. Distinguish NPCs with colour, headwear and carried props attached to bones.

Environment and props

- Kenney city and food kits, Quaternius town and nature packs, KayKit city builder: all CC0, consistent low-poly style.
- [Poly Pizza](#) to search one-off props (bowls, crates, bicycles, stalls). Check the licence on each model page; filter for CC0.
- OpenGameArt and Sketchfab have useful pieces, but licences vary per file (CC0, CC-BY, CC-BY-SA). Avoid CC-BY-SA.
- A broader list: [awesome-cc0](#).
- Chinese street dressing (lanterns, shop fronts, signboards, steamers) will be thin in free packs. Plan to model 10–15 simple props in Blender.

Asset rules

- GLB only in the build. Convert everything in Blender; strip textures to flat colours; compress with gltf-transform (Draco or Meshopt).
- Keep a `CREDITS.md` with source, author and licence for every asset, including CC0 ones.
- Stay in one art family. Do not mix realistic and low-poly models.

Reference reading on how Messenger was made: [Communication Arts write-up](#).

Technical architecture

A static site: three.js for the world, plain DOM for UI, all content in JSON, saves in the browser. No backend in Phase 1.

Stack

- TypeScript + Vite + three.js. No game engine on top.
- UI (bubbles, notebook, wallet) as HTML/CSS over the canvas, not drawn in WebGL. Hanzi render sharply and the UI is easy to change.
- `GLTFLoader` with Draco or Meshopt; `AnimationMixer` on the shared skeleton.
- Movement: click or tap to walk on a navmesh or a simple walkable grid. No physics engine.
- State: one plain store (Zustand or a hand-rolled one) for wallet, day, word states, scene flags.
- Saves: `localStorage` as versioned JSON, with export/import as a text string so testers can move devices.
- Hosting: any static host. Target under 15 MB first load, 30 fps on a mid-range Android phone.

Modules

```

flowchart TD
    C[Content JSON<br/>scenes, words, world] --> D[Dialogue runner]
    C --> W[World loader]
    L[Learner model<br/>word states] --> D
    D --> L
    D --> E[Economy<br/>wallet, day slots]
    D --> U[UI layer<br/>bubbles, notebook]
    W --> R[three.js renderer<br/>toon + outline]
    A[Audio player] --> U
    D --> A
    S[Save/load] --> L
    S --> E
  
```

The dialogue runner is the centre. It reads a scene, asks the learner model which words to put in the open slots, plays lines, takes the reply, and reports results to the learner model and the economy.

Content format

Three files per phase: `words.json` (the HSK list plus bonus words), `scenes.json`, `world.json` (locations, NPC placements, signage text). A scene looks like this:

```

{
  "id": "p1_noodle_order_01",
  "phase": 1,
  "location": "noodle_shop",
  "npc": "cook",
  "requires": ["茶", "桌子", "号"],
  "introduces": ["杯子"],
  "exchanges": [
    {
      "line": {
        "hanzi": "{n}号桌子, {count}杯茶。",
        "pinyin": "{n} hào zhuōzi, {count} bēi chá.",
        "en": "Table {n}, {count} cups of tea.",
        "audio": "p1_noodle_order_01_a",
        "words": ["号", "桌子", "杯子", "茶"]
      },
      "slots": { "n": "number_1_10", "count": "number_1_10" },
      "replies": [
        { "hanzi": "好。", "action": "serve", "check": "count_and_table" }
      ],
      "onWrong": "p1_noodle_wrong_table"
    }
  ]
}

```

Every line carries its own word tags. That is what makes the level checker and the learner model possible.

Level checker (build this early)

A script run in CI and before every content commit. It fails the build if:

- any line uses a word above the scene's phase that is not in `introduces` or marked bonus;
- an exchange introduces more than 2 new words;
- a line's `words` tags do not match its hanzi after segmentation (use jieba or a fixed dictionary segmenter);
- any word on the phase list appears in fewer than 3 scenes;
- a line is missing pinyin, English or audio.

With this in place, scenes can be drafted quickly (by a person or with an LLM) and still be trusted.

Audio

- Pre-generated audio files, not the browser's speech synthesis. Mandarin voice quality in browsers varies widely by device.
- Generate with a TTS service at build time from `scenes.json`; one file per line, plus per-word clips for the notebook. Slot lines need one clip per slot value, or clips stitched at slot boundaries; decide during the vertical slice.
- Two or three distinct voices so NPCs sound different. Opus or AAC, mono, about 32 kbps.
- Have a native speaker spot-check the output, especially tones on 一 and 不.

Text

- Bundle a subset CJK font covering only the characters used (a few hundred glyphs in Phase 1) to keep downloads small.
- Pinyin with tone marks everywhere, never tone numbers.

Build order and milestones

Prove the loop with grey boxes first; spend on art only after testers say the loop works. Week counts assume one developer part-time and are estimates to revise after M1.

Milestone	What exists at the end	Rough effort
M0 — Scene list	Spreadsheet of ~25 scenes covering all 150 words, each word in 3+ scenes; owner signs off	1 week
M1 — Vertical slice	One location (noodle shop), one job, 3 scenes, grey-box art, mannequin character, dialogue runner, wallet, audio for those lines	2 weeks
M2 — Pipeline	<code>words.json</code> for HSK 1, level checker in CI, TTS build script, save/load	1 week
M3 — Full Phase 1 in grey box	Whole street walkable, all scenes, three jobs, day cycle, rent, notebook, mentor, hidden repetition	3–4 weeks
M4 — First playtest	3–5 testers on the grey-box build; fix what confuses or bores them	1 week

Milestone	What exists at the end	Rough effort
M5 — Art pass	Toon shader, outlines, final models, signage, UI styling, music and ambient sound	2–3 weeks
M6 — Validation playtest	5–10 testers, measurements from the Validation section, go/no-go on Phase 2	2 weeks

Rules for the order:

- M1 is the decision point for the tech choices flagged as open (outline method, slot audio, navmesh vs grid).
- Nothing from M5 starts before M4 feedback is in.
- Content is written against the level checker from M2 onward; no unchecked scenes in the build.

Validation

Phase 2 gets built only if Phase 1 shows that people finish it, keep the words, and want more. The thresholds below are starting proposals for the owner to confirm before M6.

Testers: 5–10 people. A few true beginners, a few at HSK 1–2. Unsupervised play on their own device.

Question	How it is measured	Proposed bar
Do they finish without being pushed?	Share of testers reaching the Phase 2 gate within two weeks	At least 60%
Do the words stick?	30-word recognition check (audio → meaning, hanzi → meaning) 3–5 days after finishing	Beginners score 70% or more
Is it competitive with drilling?	Same check against a tester who spent equal time on a flashcard or drill site	Within 15 points, with better listening scores
Do they want more?	Asked at the end: would you play Phase 2?	Most say yes unprompted

Question	How it is measured	Proposed bar
Where does it drag?	Per-scene drop-off, replays, pinyin taps, time per exchange	No scene loses more than 20% of players

Instrumentation for this is local only: the game writes an event log into the save, and testers export it as a text string. No analytics service, no accounts.

The game will teach fewer words per minute than drilling. What it should win on is listening at natural speed, recall in context, and people coming back on their own. If it does not win on those, stop.

Open questions

The implementer should get answers to these from the owner before M1 ends.

- ☐ **No-face standard:** is a blank, featureless head acceptable, or is something stricter needed (for example hooded or covered heads for all characters)?
- ☐ **Audience:** adult self-learners, children, or both? This sets text size, humour and how much English the mentor uses.
- ☐ **Explanation language:** English only in Phase 1, or Bengali as well?
- ☐ **Title:** "Make It in China" is a placeholder.
- ☐ **Protagonist:** fixed character or player-chosen outfit and colour? Any backstory (where they came from, why they are there)?
- ☐ **City:** a generic fictional Chinese city, or modelled on a real one?
- ☐ **TTS provider and budget** for pre-generated audio, and who does the native-speaker check.
- ☐ **Validation thresholds:** confirm or change the proposed bars.
- ☐ **Animals:** do the faceless rules apply to cats, dogs and birds in the street, or should animals be left out?
- ☐ **Release intent:** private test only, or public and free, or paid later? This affects licence choices (Mixamo terms, fonts, TTS voice rights).